

A Tiny Reproducible Example Manuscript

Example source file for teaching Quarto manuscript features

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This short manuscript demonstrates the Quarto features introduced in the teaching handout: YAML metadata, Markdown structure, equations, tables, figures, cross-references, citations, executable code chunks, and multi-format output from one plain-text source file.

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1 Introduction

Quarto manuscripts combine ordinary Markdown writing with features that are useful in scientific and technical documents. The source file remains readable as plain text, while Quarto can render the same document to HTML, PDF, or Word (Posit Software, PBC 2026; MacFarlane 2026).

This example uses a small teaching dataset to show how narrative text, code, tables, figures, equations, citations, and cross-references can live in one document. The goal is not to perform a serious analysis; the goal is to show the workflow.

2 Methods

Suppose a class measures a simple score before and after a short learning activity. The observed values are shown in Table 1.

Table 1: Example observations used by the code chunk.

Group	Baseline score	Follow-up score
A	62	71
B	67	74
C	59	68
D	73	80
E	65	77

For each group, the change score is defined in Equation 1.

$$\Delta_i = y_{i,\text{followup}} - y_{i,\text{baseline}} \quad (1)$$

The analysis below shows executable Quarto code syntax. Evaluation is disabled in the YAML header so the file renders on classroom machines without requiring a Python/Jupyter setup; set `execute.enabled` and `execute.eval` to `true` after installing those tools to run the chunks during rendering.

3 Results

The following figure summarizes the change scores from Table 1.

The executable Python chunk below creates the same values shown in Table 1 and computes the change score from Equation 1.

```
groups = ["A", "B", "C", "D", "E"]
baseline = [62, 67, 59, 73, 65]
follow_up = [71, 74, 68, 80, 77]
change = [after - before for before, after in zip(baseline, follow_up)]
```

The mean change score is computed in the next executable chunk.

All five groups improved in this small example. The average increase was 8.8 points. When execution is enabled, the chunk above can regenerate the value from the source data instead of relying on manual copy and paste.

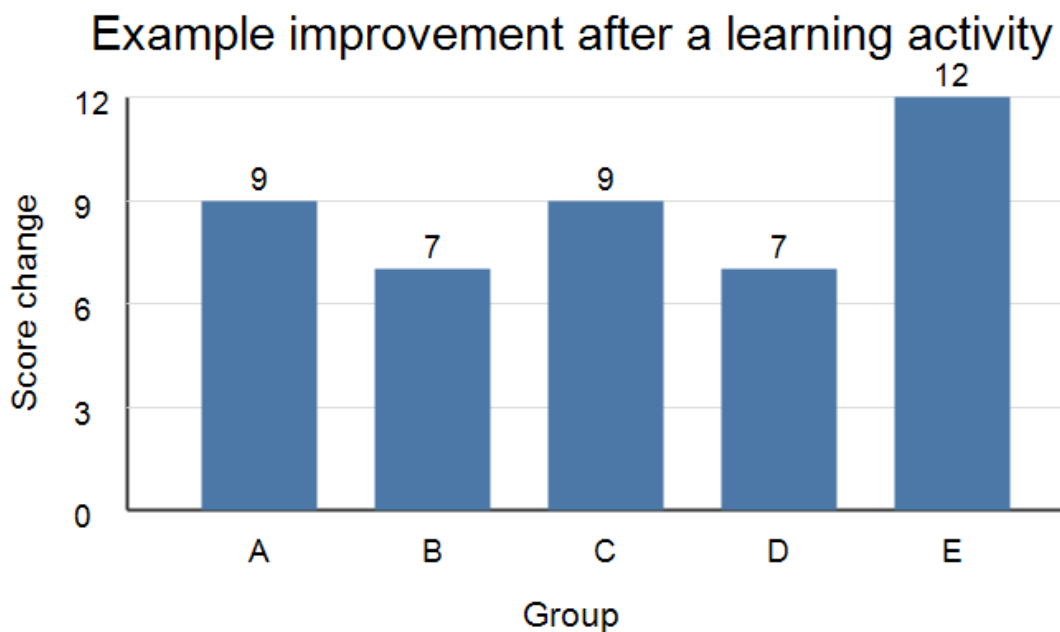


Figure 1: Change from baseline to follow-up for five example groups.

Listing 1

```
mean_change = sum(change) / len(change)
print(f"Mean change score: {mean_change:.1f} points")
```

4 Discussion

This manuscript demonstrates several Quarto habits that are useful for student projects:

- Use headings to create a clear document structure.
- Put metadata and output options in the YAML header.
- Label important objects such as Section 2, Table 1, Equation 1, and Figure 1.
- Keep code and explanation close together so results can be regenerated.
- Cite sources using citation keys such as [quarto2026].
- Render the same source file to multiple output formats.

For example, the same source file can be rendered with:

```
quarto render quarto_section6_example_manuscript.qmd --to html
quarto render quarto_section6_example_manuscript.qmd --to docx
quarto render quarto_section6_example_manuscript.qmd --to pdf
```

References

MacFarlane, John. 2026. *Pandoc: A Universal Document Converter*. <https://pandoc.org/>.

Posit Software, PBC. 2026. *Quarto: An Open-Source Scientific and Technical Publishing System*. <https://quarto.org/>.